



Public Transit User Dynamics: A Simulation Framework Under Large-Scale Event in [location] Context

Damar Pratita Dewayanto
D255113

Supervisor: Prof. Makoto Chikaraishi

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General problem

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Research objective

Research question

Research gap and contribution

Future task

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General Problem

- Modern cities have increasingly promoted the use of public transportation.
- This trend has led to a growing demand for public transport systems, particularly metro rail and city bus services, within urban contexts.
- Instead of non-peak hour, passengers demand intensifies during peak hour serving various trip purposes such as commuting to work and school.

General Problem

- Large-scale event
 - While peak hour already challenges public transport operations under normal condition, these issues can be significantly intensified under atypical conditions.
 - Such uncommon conditions are often observed during the large-scale events.
 - Large-scale event means event that draw substantial numbers of individuals to a location, placing the local environment and infrastructure under great pressure, and bringing disruption to residents (Parkes S.D., et.al.,2016).
 - Such phenomena has been conceptualized in the literature including [mega-events](#) (Parkes S.D., et.al.,2016), [disruption/service disruption](#) (Pnevmatikou et.al,2015; Rahimi et.al.,2019), [shock to the transport system](#) (Jenelius E., and Mattsson, L.G.,2021), and [large-scale event](#) (Kopsacheilis A., 2026)
 - Some of them, such as the London Olympic Games and Paralympic Games 2012 (Parkes S.D., et.al.,2016) and FIFA World Cup Qatar 2022 (Mohammed, A., et.al.,2025), reported unusual conditions in public transport operation (e.g., sky-rocketing ridership during the event, additional significant loading to the station that already overcapacity).
 - Accordingly, critical processes such as passenger movement/dynamics within boarding and alighting operations can serve as key contributors to these delays

General Problem

- Microscopic pedestrian modeling
 - Pedestrian dynamics models could be classified into two primary categories namely **continuous model** and **discrete-based model** (Chen,Xu, et.al., 2017).
 - Among the continuous approaches, the Social Force Model (SFM), originally proposed by Helbing, D., and Molnar, P.,(1995), remains one of popular frameworks for simulating pedestrian dynamics.
 - SFM has been explored to enhance the understanding of pedestrian dynamics across various context, including in open space such as sidewalks (Siddhart, S.M.P. and Vedagiri, P., 2018, Zhao X., et.al., 2026), emergency situation (Zhang, et.al.,2023), and metro transit system, specifically during boarding and alighting process (Yining Jia, et.al.,2026).

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Rationale

- Although boarding-alighting assessment studies have gained attention, since this critical process can affect public transport performance, most prior studies have primarily focused on passenger movements associated with entering and exiting vehicles during boarding and alighting processes (Qu Y., et.al.,2019, Seriani and Fernandez, 2015).
- At this moment, only Yining Jia, et.al.(2026) explicitly consider a categorization of metro passengers into three groups: **boarding**, **alighting**, and **remaining on-board passengers**.
- However, their analysis is situated within normal operating conditions, where remaining passengers on-board are typically seated, using experimental settings by Fu L., et.al., (2023).
- This study emphasizes the dynamics of remaining on-board passengers, as part of adaptive mechanism under crowding conditions induced by large-scale events.
- Understanding these interactions is critical, as adjustments made by on-board passengers can significantly influence passenger flow efficiency, dwell time, and overall operational performance.
- Notably, this context has received limited attention within the Social Force Model (SFM) framework.
- In addition to complementing continuous approach, this study will consider discrete behavioral modeling approach, introduced by Robin, Th., et.al.,(2009)

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Research Objective

This study aims to examine the public transit user adapting behavior under the large-scale event-induced crowding situation

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Research Objective

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How do the adaptive behaviors of passengers influence flow dynamics and operational performance (e.g., dwell time and congestion) during boarding and alighting processes under large-scale event-induced crowding conditions?

2

tbd

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Research Gap and Contribution

Type of gap	Identified gap	Study approach	Academic contribution	Practical contribution
Empirical gap	Lack of empirical or simulation-based studies on passenger dynamics during the large-scale event-induced crowding situated on public transport system	Examines passenger dynamics under high-density conditions	Provides new evidence on passenger behavior and system performance in disrupted scenarios	By understanding how remaining on-board passengers adapt (e.g., repositioning, yielding space), operators can design better boarding/alighting procedures or crowd management during the operations
Theoretical gap	The SFM emphasizes individual movement/dynamics by forces but underrepresent on adaptive behavior on remaining passengers during the boarding-alighting process	Introduces adaptation mechanism of on-board passengers within pedestrian dynamics framework	Extends SFM by incorporating adaptive behavior in high density context	<i>tbd</i>

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Future Task

- Decide the large-scale event type context which taking to account
- Expand understanding on discrete-based approach
- After the fundamental principal confirmed with Sensei, discuss and decide data collection method

References

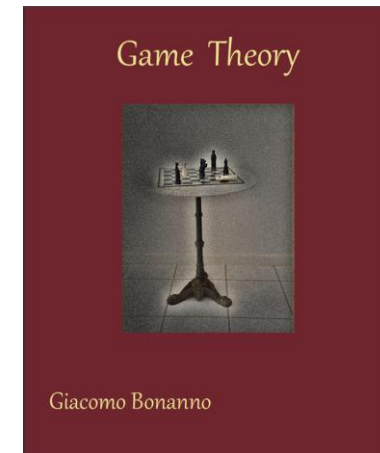
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Appreciate your feedback and discussion :D

Upcoming Lab Activity

Upcoming Lab Activity

- proposed date:
 - July 17-20th (concern: long weekend -> less course schedule, after master thesis manuscript submission to Sensei July 15, to IDEC July 31, but conflicting to ICMC conference)
 - July 24-26th (concern: one week before final exam)
 - August 11-14th (concern: after TSE and SmaSo thesis defense, beginning of the summer holiday)
- proposed location and venue:
 - Okayama Prefecture, Kurashiki
 - Shimane Prefecture, Izumo
 - Hiroshima Prefecture, Etajima
- proposed reading list (reference) and topics:
 - 1) Title “Game Theory”:
[https://faculty.econ.ucdavis.edu/faculty/bonanno/PDF/GT_book.pdf#:~:text=This%20book%20is%20suitable%20for,\(in%20its%20entirety\)%20a%20first%2Dyear](https://faculty.econ.ucdavis.edu/faculty/bonanno/PDF/GT_book.pdf#:~:text=This%20book%20is%20suitable%20for,(in%20its%20entirety)%20a%20first%2Dyear)
 - 2) Title “An Introduction to Game Theory”:
https://mathematicalolympiads.wordpress.com/wp-content/uploads/2012/08/martin_j-osborne-an_introduction_to_game_theory-oxford_university_press_usa2003.pdf



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